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2172 BEDVH: A Novel Method for Evaluating Biologic Effective Dose for Eye Plaque Brachytherapy Implants

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Purpose/Objective(s): Eye plaque brachytherapy, like most brachytherapy, is characterized by very heterogeneous dose distributions. The biologic effective dose (BED) of brachytherapy treatments depends heavily upon the duration of the implant and the α/β ratio of the treated and normal tissues. We propose a new method to evaluate eye plaque brachytherapy implants based upon the BED delivered to the tumor and surrounding normal ocular structure volumes.

Materials/Methods: Three dimensional CT scan was used to contour uveal melanoma tumor volume and adjacent normal structures including the lens, optic nerve, and lacrimal gland. Monte Carlo methods were used to calculate tissue heterogeneity-corrected physical dose delivered to the tumor and these structures with a 16 mm ^{125}I COMS eye plaque for a dose of 85 Gy prescribed to 5 mm. High resolution dose volume histogram (DVH) data were exported directly from Pinnacle TPS. Using the equation for temporary low dose rate brachytherapy implants introduced by Dale (*British Journal of Radiology*, 1989), physical DVH data were then converted to BED-volume histogram (BEDVH) data using the assumptions previously outlined by our group (Leonard *et al.*, *Brachytherapy*, 2011). $\alpha/\beta_{\text{tumor}}$, $\alpha/\beta_{\text{lacrimal gland}}$, $\alpha/\beta_{\text{optic nerve}}$, and $\alpha/\beta_{\text{lens}}$ were assumed to be 3.5-14.1 Gy, 2 Gy, 1.5 Gy, and 1.2 Gy, respectively, based on values presented in the literature. BED was calculated for 2-7 day implants and BEDVHs were then constructed and compared to the physical dose DVH.

Results: Both the slope and scale of the BEDVH curves are highly dependent upon implant duration. Table 1 outlines the physical dose, maximum BED (BEDmax), BED_{10%}, BED_{50%}, BED_{90%}, and BED_{95%}. For normal ocular structures, BEDmax is a factor of 1.4-2 times greater for a 2 day implant than for a 7 day implant. BED_{95%}, however, does not change significantly with implant duration for the lacrimal gland and optic nerve, but increases by a factor of 1.5 for the lens. BED_{95%tumor} increases by a factor of 2.7 and BED_{95%tumor} by a factor of 1.8 for a 2 day implant as compared to a 7 day implant.

Conclusions: We have introduced a method to analyze BED delivered to any delineated structure receiving dose from a temporary low dose rate eye plaque brachytherapy implant. This technique could be used to determine and evaluate any temporary or permanent brachytherapy implant in a variety of clinical situations.

BEDVH Data: Tumor and normal ocular structures

	Physical Dmax (Gy)	BEDmax (Gy)		BED 10% (Gy)		BED 50% (Gy)		BED 95% (Gy)	
		2 day	7 day	2 day	7 day	2 day	7 day	2 day	7 day
Tumor ($\alpha/\beta = 3.5$ Gy)	326	2952	1102	1243	511	449	214	191	206
Tumor ($\alpha/\beta = 14.1$ Gy)	326	972	516	739	412	196	139	100	76
Lacrimal Gland ($\alpha/\beta = 2$ Gy)	15	25	18	19	14	9	8	0.1	0.1
Optic Nerve ($\alpha/\beta = 1.5$ Gy)	51	177	88	83	47	41	27	8	7
Lens ($\alpha/\beta = 1.2$ Gy)	32	104	53	75	41	49	29	31	20

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2173 Factors Predictive of Developing Radiation Retinopathy (RR) and Vision Loss following Plaque Brachytherapy for Uveal Melanoma

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Purpose/Objective(s): To determine factors predictive of radiation retinopathy (RR) and visual loss (20/200 or worse) following I-125 and Ru-106 plaque brachytherapy for uveal melanoma.

Materials/Methods: All patients treated at the Cleveland Clinic for a primary single unilateral ciliary body or choroid melanoma between 1/1/2005 and 6/30/2010 with I-125 or Ru-106 plaque brachytherapy were included. Patients treated for iris melanomas were excluded from analysis. Patients with bilateral melanomas or who had been previously treated with other methods, including transpupillary thermotherapy, were also excluded. Primary endpoints were loss of visual acuity and development of RR. Factors analyzed were gender, age, history of cataracts, history of diabetes, tumor size (basal dimension and apical height), tumor location, and radiation dose to the apex, fovea, and optic disc. Multivariate and univariate Cox proportional hazards were used to determine the influence of baseline patient factors on the development of RR or vision loss. Kaplan-Meier curves (log rank analysis) were used to estimate freedom from RR and vision loss.

Results: One hundred eighty-nine patients were eligible, of which 174 were still alive as of February 1, 2011. Of the 15 deaths, 4 were related to uveal melanoma, 5 were unrelated, and 6 were due to unknown causes. 6 patients required enucleation for continued tumor growth and 1 required enucleation due to painful glaucoma. 12 patients developed metastases to the liver, with 1 also developing lung metastases. 53 developed RR, 116 had post-treatment visual acuity loss $\leq 20/50$ and 58 had post-treatment visual acuity loss $\leq 20/200$. The only post treatment related side effects were RR or loss of visual acuity. No patients required additional laser treatments, developed ocular infections, or required re-admissions within 30 days of the procedure. On multivariate analysis, increased age (HR of 0.97[0.95-0.99], $p = 0.003$) and distance to the optic disc